		Coding Iteration				
ID	**Idea**	**1**	**2**	**3**	**4**	**5**
Seed 1	e-fold	+ 818 (13.1%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Seed 2	Adaptive Learning Rate	+ 95 (1.5%)	+ 766 (12.1%)	+ 407 (5.7%)	+ 28 (0.4%)	+ 1 (0%)
1	Factor Pruning	Failed to run				
2	Dynamic Factor Adjustment	+ 538 (8.6%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
3	Green Metrics	+ 168 (2.7%)	+ 97 (1.5%)	+ 162 (2.5%)	+ 176 (2.6%)	-18 (0.3%)
4	Quantized SGD	+ 1173 (18.7%)	-36 (0.5%)	+ 17 (0.2%)	+ 258 (3.5%)	+ 169 (2.2%)
5	Time Aware Early Stopping	+ 392 (6.3%)	+ 4 (0.1%)	0 (0%)	0 (0%)	0 (0%)
6	Interaction Priority	Failed to run				
7	Micro Batching	Failed to run				
8	Sparse Aware SGD	+ 524 (8.4%)	+ 1 (0%)	0 (0%)	0 (0%)	0 (0%)
9	Cluster Aware SGD	Failed to run				
10	Hybrid MF	Failed to run				

Table 1. Net change in character count (sum of additions and deletions) compared to the previous iteration. The baseline implementation contained 6,260 characters across 225 lines of code.

ideally, comparable performance) [1, 4, 9]. There are many more similar examples that are available in our GitHub repository.

These issues reveal a fundamental limitation: the AI Scientist cannot critically assess its own results. It fails to detect methodological flaws or logical inconsistencies, making it unsuitable for autonomous scientific inquiry. Without human oversight, its findings could mislead researchers.

2.5 Manuscripts

The AI Scientist generated manuscripts for all seven successfully conducted experiments, with lengths ranging from six to eight pages (median: seven). Most manuscripts (6 of 7) included tables or charts, though quality varied. References were scarce, with a median of five references per paper (range: 2–9), mostly outdated; only five (14.7%) of the total 34 references were from 2020 or later. Foundational works, such as Goodfellow et al.'s *Deep Learning* textbook, were frequently cited, suggesting weak retrieval of relevant literature.

Structural issues were common: four manuscripts (57%)contained missing or misplaced figures, incomplete sections (e.g., "Conclusions Here" placeholder), duplicates, or repeated figures. Related work sections were particularly poor in all manuscripts, often with irrelevant citations. In our e-fold cross-validation experiment, the AI Scientist failed to cite existing papers on the topic that used exactly the same terminology (e-fold cross validation), despite their availability on Semantic Scholar[17].

Results sections were often inadequate or misleading. Metrics were presented in plain text or simple charts, with no confidence intervals or p-values. Four of seven manuscripts (57%) contained incorrect or hallucinated numerical results, with discrepancies in hyperparameters and performance metrics. In one case, the AI Scientist claimed to have successfully developed an energy-efficient machine learning algorithm, supposedly evidenced by better RMSE and reduction of training time from 116 to 115 seconds – but increased memory usage. How an algorithm that is optimized for energy usage (or training time) may lead to better RMSE was not discussed (one would expect the opposite). Neither was discussed if the one-second-improvement (or 1%), in training time was statistically (or practically) significant;

[17] https://www.semanticscholar.org/search?q=%22e-fold%20cross%20validation

nor was the implication discussed that memory usage increased. Similar logical inconsistencies appeared in other experiments.

Given these flaws, these manuscripts would likely be rejected at reputable conferences or journals. Even at lower-tier venues, the lack of proper citations and unreliable results would be problematic. However, an undergraduate submitting such work for a course project might still pass if the instructor does not scrutinize the code or experiments. Therefore, in our opinion, the AI Scientist's output resembles that of a student rushing to complete an assignment – formally structured but careless in execution.

2.6 Review Functionality

The AI Scientist includes a reviewer agent for AI-assisted peer review, evaluated by us on both AI-generated and human-written manuscripts. To generate a review, the AI Scientist extracts text from a provided PDF but cannot interpret figures, tables, or supplementary material. It outputs a structured review (`review.txt`) containing a summary, identified strengths and weaknesses, improvement questions, numerical ratings for originality, clarity, significance, soundness, presentation, and contribution, plus a binary accept/reject recommendation.

Reviews of AI-Generated Manuscripts. For all seven manuscripts generated by the AI Scientist, its own review system recommended rejection. One common critique was that the research was based on a single, small dataset (MovieLens 100k). This criticism is justified, and the use of the single dataset not an intrinsic flaw of the AI Scientist itself, but rather a consequence of our experimental pipeline, which only included this dataset. Had our pipeline contained multiple datasets, the AI Scientist likely would have incorporated them. That being said, in an ideal scenario, the AI Scientist would be capable of retrieving and utilizing additional datasets autonomously, rather than being entirely dependent on user-supplied input.

The AI Scientist highlighted more valid concerns, such as weak theoretical justification and potential biases, but overlooked critical flaws, including redundant text, formatting errors, missing sections, and flawed experimental results. Most notably, the reviewer agent did not identify any of the truly serious issues that we highlighted in the sections above.

While human reviews vary in quality – from superficial comments to detailed critiques – the AI Scientist's reviews were consistently structured, though lacking depth. Despite this, they appeared plausible enough that editors and conference chairs might not immediately recognize them as AI-generated. While less rigorous than strong human reviews, they were still more substantive than the brief, low-effort reviews that sometimes occur in the conference reviewing process.

Reviews of Human-Written Manuscripts. We tested[18] the AI Scientist's reviewing capabilities with ten human-written papers on OpenReview.net[19], five that were accepted and five rejected. Reviews created by the AI Scientist followed the same structured format as for AI-generated manuscripts, including summaries, critiques, improvement suggestions, and numerical ratings.

While indicative, the results were striking: the AI Scientist rejected 9 out of 10 papers, including four that had been accepted by human reviewers, while recommending acceptance for only one, which had been rejected on OpenReview. This suggests a strong conservative bias and misalignment with human judgment. Despite this, its reviews raised valid concerns about methodology and biases but lacked depth, often missing broader contributions and argumentation.

[18] To accomplish this, we modified the AI Scientist; this modification is available on our GitHub account.
[19] https://openreview.net/

Our evaluation of the AI Scientist reviewer agent, illustrates a challenge: how to evaluate AI-generated reviews, given the inconsistencies of human peer review? Should human judgment be the benchmark? Peer review is known for subjectivity, reviewer disagreements, and biases. An AI system that diverges from human reviewers may still highlight valid weaknesses or apply overly rigid criteria.

This raises another fundamental question: should AI peer review aim to mimic human reviewers or provide a more systematic alternative? If the former, the AI Scientist falls short. If the latter, human evaluation alone may be an insufficient benchmark. A better assessment would compare AI-generated reviews against expert evaluations that measure consistency and validity.

Ultimately, while structured and plausible, the AI Scientist's reviews lack the depth needed for academic evaluation. It follows predefined templates but struggles with context-dependent judgment. Without improvements in assessing contributions, argumentation, and methodology, it remains better utilized as feedback tool, rather than a peer-review replacement.

2.7 Costs and Effort

When assessing the AI Scientist's potential, cost is a crucial factor — and surprisingly, it remains low. The total cost[20] for our main experiments, which included generating 10 ideas, running experiments for 7 ideas (from 12 total, including 2 seed ideas, of which 5 failed), and producing 7 manuscripts with reviews, amounted to only $42 USD — an average of $6 per manuscript. Given that our experiments were less complex than typical recommender systems research or those conducted by Sakana, their claim that a full research paper can be produced for around $15 seems realistic.

However, human labor is still required. Setting up the AI Scientist took approximately 5 hours, while implementing the experimental template required an additional 15 hours. It must be noted that if we were to re-implement similar pipelines, the required time would decrease due to the learning curve. If we were to implement more complex experiments, however, time would increase. Beyond this, we spent about 10 more hours brainstorming initial seed ideas, reviewing outputs, and refining results. Excluding the initial setup time, this totals 25 hours of human effort — approximately 3.5 hours per manuscript — mainly from an undergraduate student.

In summary, producing a research paper — even of relatively low quality — at an average cost of $6 to $15 and 3.5 hours of human effort, is remarkable. By comparison, we estimate that an undergraduate student would require at least 20 to 40 hours, while an experienced researcher — who would likely be reluctant to produce work of similar quality — might take 10 to 20 hours. Thus, the AI Scientist operates approximately 3 to 11 times faster than a human researcher at negligible costs.

2.8 Limitations

Our own evaluation had limitations: we used a single dataset (MovieLens-100k) and one research domain (Green Recommender Systems), relied on fixed seed ideas and one template. While these constraints ensured a controlled assessment, they may affect generalizability and reproducibility. Despite this, we believe our findings remain meaningful.

[20] We report only costs of OpenAI's API; other costs such as electricity for our laptop or university cluster are ignored